15 *A*, *B*, *C* and *D* are four points on a straight line.



$AB : BD = 1 : 2$
$AC : CD = 5 : 7$

Show that $BD = 8BC$

Choose AD = 12 equal units.

AB : BD = 1 : 2 = 4 : 8

1 mark
Match both whole lengths to 12.

AC : CD = 5 : 7

BC = AC – AB = 5 – 4 = 1 unit

1 mark
Subtract to find BC.

BD = 8 units = 8BC, as required.

1 mark
The common unit cancels; this proves the relationship.

(Total for Question 15 is 3 marks)

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